

DATABASE

CAT1

MARKING SCHEME

SECTION A: 20 MARKS

Question 1: Define the following terms (4 marks)

a) Database – 1 mark

An organized collection of related data stored and managed electronically for easy access, retrieval and management. **(1)**

b) Database Management System (DBMS) – 1 mark

Software used to create, store, organize, retrieve, update and manage data in databases. **(1)**

c) Data redundancy – 1 mark

The unnecessary or repeated duplication of the same data in a database or data storage system. **(1)**

d) Data independence – 1 mark

The ability to change the database structure/schema at one level without requiring changes at another level or in application programs. **(1)**

Total: 4 marks

Question 2: State four functions of a DBMS (4 marks)

Award **1 mark each** for any four correctly stated functions:

- Data storage and organization **(1)**
- Data retrieval/query processing **(1)**
- Data insertion, updating and deletion **(1)**
- Security and access control **(1)**
- Backup and recovery **(1)**
- Integrity/constraint enforcement **(1)**
- Transaction management **(1)**
- Concurrency control **(1)**
- Data definition/schema management **(1)**

Maximum: 4 marks

Question 3: Application areas of database systems (4 marks)

Award **1 mark for each correct application area with a relevant example.**

Application area	Example	Marks
Banking/Finance	Customer accounts, transactions and loans	1
Education	Student records, courses and examination results	1
Healthcare	Patient records, appointments and medical history	1

Retail/Business	Products, customers, sales and inventory	1
Library	Books, members and borrowing records	1
Airline/Transport	Passenger bookings and flight schedules	1

Maximum: 4 marks

Question 4: Centralized and Distributed DBMS (4 marks)

a) Centralized DBMS

Definition/distinction – 1 mark:

A DBMS in which the database is stored and managed at a **single central location/server**, with users accessing the central database. **(1)**

Example – 1 mark:

A college student database hosted on one central server accessed by authorized departments. **(1)**

b) Distributed DBMS

Definition/distinction – 1 mark:

A DBMS in which the database/data is distributed across **multiple networked locations or sites**, while being managed as a coordinated database system. **(1)**

Example – 1 mark:

A bank database system where data is maintained across several geographically separated branches/sites. **(1)**

Total: 4 marks

Question 5: Advantages of a DBMS over a traditional file-based system (4 marks)

Award **2 marks for each correctly explained advantage**, up to four marks.

Possible answers:

1. Reduced data redundancy – 2 marks

A DBMS can reduce unnecessary duplication of data by centrally managing shared data. **(2)**

2. Improved data security – 2 marks

A DBMS provides authentication, authorization and access controls to restrict unauthorized access. **(2)**

3. Improved data integrity – 2 marks

Constraints and validation rules help ensure that stored data remains accurate and consistent. **(2)**

4. Data sharing – 2 marks

Multiple authorized users and applications can access and share the same database. **(2)**

5. **Backup and recovery – 2 marks**

DBMS facilities can support backup and recovery following failures or data loss. (2)

6. **Data independence – 2 marks**

Changes to the database structure can often be made without requiring major changes to application programs. (2)

Maximum: 4 marks

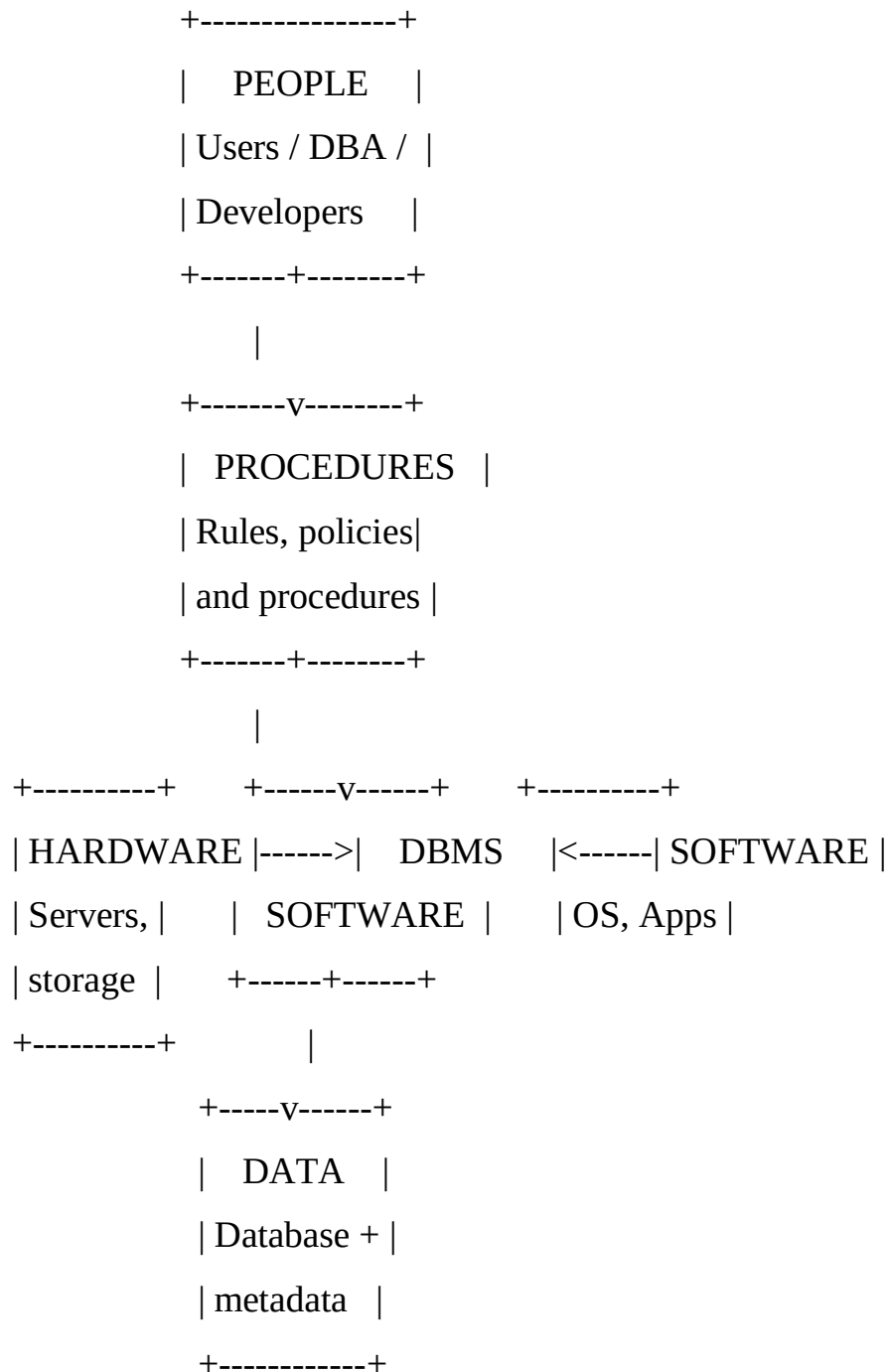
SECTION B: 20 MARKS

Question 6: Five major components of a DBMS environment (10 marks)

Expected diagram

The candidate should provide a **well-labelled diagram** showing the major components. A suitable representation is:

diagram may show:



Mark allocation

Award **2 marks** for each correctly identified and described component.

1. Hardware – 2 marks

- Physical equipment used to operate and store the database. **(1)**
- Examples: servers, computers, storage devices and networking equipment. **(1)**

2. Software – 2 marks

- Programs used to operate and manage the database environment. **(1)**
- Includes DBMS software, operating systems and application software. **(1)**

3. Data – 2 marks

- The actual information stored and managed by the database. **(1)**
- Includes operational data and metadata/database descriptions. **(1)**

4. Procedures – 2 marks

- Rules and instructions governing how the database is created, used, secured, maintained and backed up. **(2)**

5. People – 2 marks

- Individuals who interact with or manage the database system. **(1)**
- Examples include DBA, database designers, developers and end users. **(1)**

Total: 10 marks

Diagram: Award the component marks even if the candidate's diagram differs from the example, provided the five major components are correctly identified and logically represented.

Question 7: Explain the three-tier database architecture (6 marks)

Award 2 marks for each tier.

1. Presentation tier – 2 marks

- The user interface through which users interact with the system. **(1)**
- It displays information and collects input from users. **(1)**

Example: Web browser or graphical user interface.

2. Application/Business Logic tier – 2 marks

- Contains the application programs and business rules that process user requests. **(1)**
- Acts as an intermediary between the presentation layer and database layer. **(1)**

Example: Application/web server.

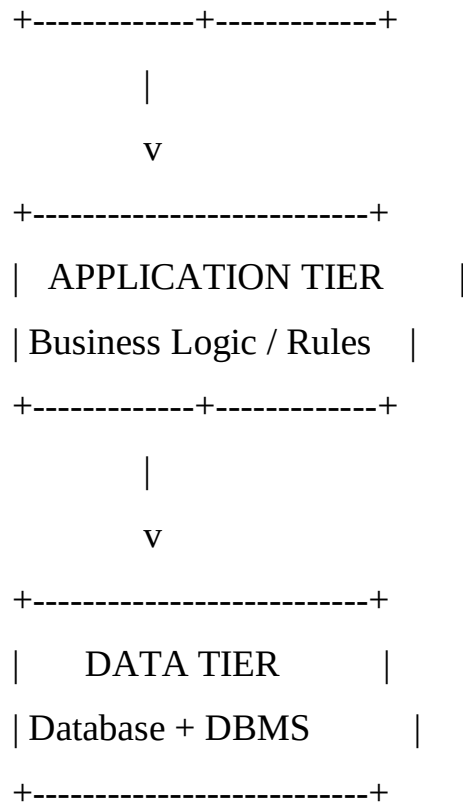
3. Data/Database tier – 2 marks

- Contains the database and DBMS that store and manage data. **(1)**
- Handles data storage, retrieval, updating and database operations. **(1)**

Example: MySQL, PostgreSQL, Microsoft SQL Server or Oracle Database.

Expected diagram





Total: 6 marks

Question 8

a) Define requirements elicitation (2 marks)

Requirements elicitation is the systematic process of **collecting, discovering and documenting the needs, expectations and requirements of users and other stakeholders** for a database system. (2)

Award:

- Collecting/discovering requirements from users/stakeholders (1)
- Correct reference to system/database needs or requirements (1)

b) Techniques for gathering database requirements (2 marks)

Award **1 mark each** for any two:

- Interviews **(1)**
- Questionnaires/surveys **(1)**
- Observation **(1)**
- Workshops/group discussions **(1)**
- Document analysis **(1)**
- Focus groups **(1)**
- Reviewing existing systems/processes **(1)**

Total: 2 marks

SECTION C: 10 MARKS

Question 9

a) Components of a database requirement assessment report (3 marks)

Award **1 mark each** for any three appropriate components.

Possible answers include:

1. **Introduction/background** – explains the organization, purpose and scope of the proposed database. **(1)**

2. **User/stakeholder requirements** – identifies users and their information needs. (1)
3. **Functional requirements** – describes what the database system should be able to do. (1)
4. **Data requirements** – identifies the data/entities and information to be stored. (1)
5. **Business rules** – documents rules and conditions that the system must enforce. (1)
6. **Security requirements** – identifies access-control and data-protection needs. (1)
7. **Current-system assessment** – describes weaknesses and limitations of the existing system. (1)
8. **Recommendations/conclusion** – proposes solutions based on the assessment. (1)

Maximum: 3 marks

9(b): Relational vs Hierarchical Database Model (7 marks)

i. Structure of data representation (2 marks)

Relational model – 1 mark

Data is organized into **tables (relations)** consisting of rows/records and columns/attributes. (1)

Hierarchical model – 1 mark

Data is organized in a **tree-like structure** consisting of parent and child records/nodes. **(1)**

ii. Relationship between records/entities (2 marks)

Relational model – 1 mark

Relationships are established using **keys**, particularly primary keys and foreign keys. It can represent one-to-one, one-to-many and, through related tables, many-to-many relationships. **(1)**

Hierarchical model – 1 mark

Records are connected through **parent-child relationships**, traditionally supporting a one-to-many structure. **(1)**

iii. Flexibility in querying data (2 marks)

Relational model – 1 mark

Highly flexible; users can use query languages such as **SQL** to retrieve and combine data from multiple related tables. **(1)**

Hierarchical model – 1 mark

Less flexible for ad-hoc queries because data is accessed mainly by following predefined paths through the hierarchy. **(1)**

iv. Real-world example (1 mark)

Award **0.5 mark each** for an appropriate example:

- **Relational model:** MySQL, PostgreSQL, Oracle Database, Microsoft SQL Server, or a typical college/student management database. **(0.5)**
- **Hierarchical model:** IBM IMS or a traditional organizational/file hierarchy. **(0.5)**

Total: 7 marks

General marking instructions

1. **Accept equivalent correct answers** expressed in different wording.
2. Do not award marks for merely giving an example where an explanation/definition is required.
3. Where a question asks for an explanation, award the mark only when the candidate demonstrates the relevant concept.
4. For Q6, award diagram marks based on **correct identification and representation**, not artistic quality.
5. For Q9(b), candidates may present the comparison in a table; award marks according to the four specified aspects.
6. **No negative marking** for incorrect answers unless institutional examination regulations specify otherwise.

7. Where more answers than required are given, mark the **first required number of valid responses**, unless your institutional policy states otherwise.